

Aneesh Venkatesha Rao

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EDUCATION

National Institute of Technology Warangal, Telangana, India

Aug 2024 — May 2028

B.Tech in Electronics and Communication Engineering

Delhi Public School - Modern Indian School (DPS-MIS), Doha, Qatar

— Apr 2024

High School Diploma (CBSE) - PCM with Computer Science

SKILLS

Programming Languages: Python, Go, TypeScript, JavaScript, C++, SQL

AI & ML: RAG, LLMs, LangChain, LangGraph, pgvector, HuggingFace, PyTorch, Whisper, Cohere, Gemini API, OpenAI API, Ollama

Backend & Databases: FastAPI, Node.js, Pydantic, PostgreSQL, asyncpg, Docker, CI/CD

Frontend & Tools: Next.js, React, Tailwind CSS, Vercel, Railway, Git, GitHub, Figma

Hardware / FPGA: Verilog, Xilinx Vivado

PROJECTS

ContextCraft | Python, FastAPI, PostgreSQL, pgvector, tree-sitter, Cohere, Next.js

Apr 2026 — May 2026

- Engineered an AST-aware codebase search engine using tree-sitter across Python, JavaScript, TypeScript, and Go, and implemented hybrid Reciprocal Rank Fusion (pgvector cosine similarity + PostgreSQL BM25); achieved **80% source hit rate** at **3.88s P50** latency across a 30-query, 3-iteration evaluation harness.
- Designed per-repo RRF normalization preventing large codebases from dominating multi-repo results; added Cohere cross encoder reranking (20 to 10 candidates), 1-hop cross-file dependency expansion with cycle detection, and published to PyPI as contextcraft-py with a FastAPI SSE backend and GitHub Actions CI enforcing mypy strict, ruff, and pytest.

Dev-Saarathi | TypeScript, Python, AWS Lambda, Bedrock, DynamoDB, S3

Jan 2026 — Mar 2026

- Built a VS Code extension enabling voice and text codebase queries in **11 Indian languages**, architecting a serverless AWS pipeline (API Gateway – Lambda – Bedrock Nova Pro) with Amazon Transcribe STT, Bedrock Knowledge Base RAG, and Bedrock Guardrails for content safety.
- Implemented an intent detection and execution router supporting autonomous agentic actions (Git commits, README generation, test scaffolding) with a human-in-the-loop approval UI maintaining **3-6s voice-to-response latency**; engineered exponential backoff polling (3s – 5s – 8s) reducing redundant Lambda invocations **~45%** and compute costs **~35%**.

ShabdSetu | Next.js, FastAPI, AI4Bharat IndicTrans2, Whisper, Firebase

Sep 2025 — Dec 2025

- Built an AI language learning platform for **10 Indian languages (1.4B+ combined speakers)** using AI4Bharat IndicTrans2 (200M parameters) with **95%+ translation accuracy** across 800+ curated phrases, a 3-tier fallback pipeline (IndicTrans2 – HuggingFace API – Google Translate), and Whisper Large V3 word-level fuzzy pronunciation scoring.
- Achieved **sub-50ms** cached response times via multi-layer caching (in-memory + Firestore) with automated cache warming and **3-10x query speedup** through 26 optimized SQLite indexes; built a gamification engine with 27 achievements across 6 tiers, XP leveling, daily streaks, and global leaderboards synchronized via Clerk JWT authentication.

AWARDS & HONORS

Smart India Hackathon 2025

December 2025

National Finalist: Selected among 1,300 teams from 75,000+ submissions; ranked Top 5 in our problem statement.

Olympiad Awards

April 2024

SOF, SilverZone, NSTSE

Earned 65+ competitive International Olympiad medals over 7 consecutive years, validating elite-level quantitative aptitude, analytical problem-solving, and logical reasoning skills.

CERTIFICATIONS

Professional Certifications

June 2025

Google, Meta, & IBM

Credential ID: Completed 13 professional certifications from Google, Meta, and IBM, mastering core methodologies in cloud architecture, software engineering pipelines, and data-driven development.